TIC 229181348

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        6.120919 d
  Epoch (t0)    1574.1034 BTJD
  Duration      0.0840 d
  Depth         13.080 %  (130,802 ppm)
  SNR           358.3

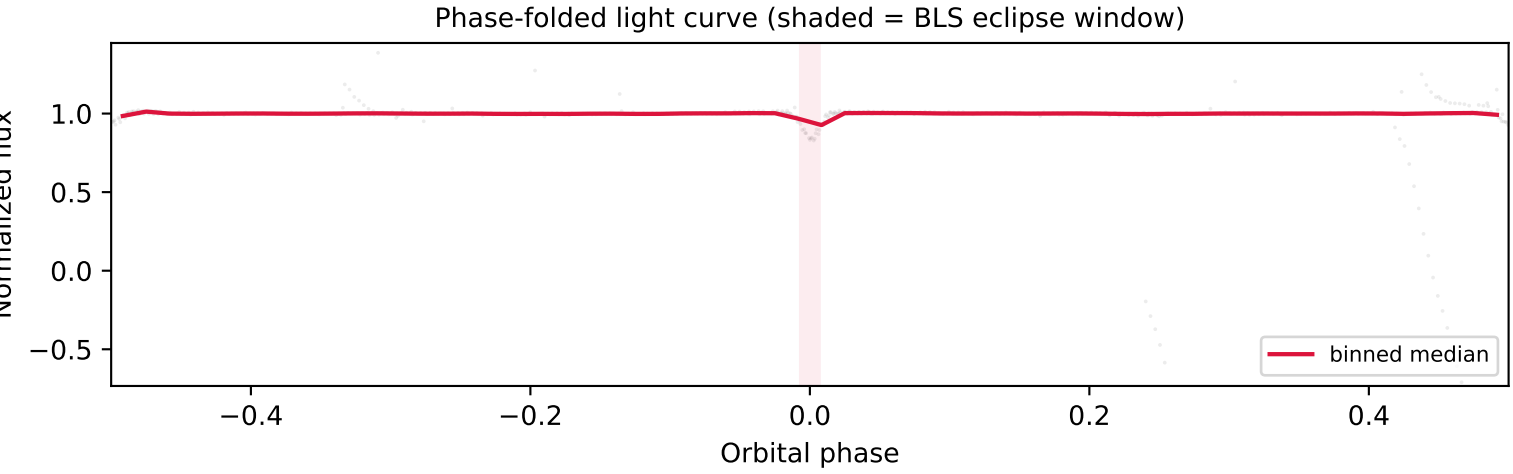
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      4  (gate >= 3)
  Robust depth ratio     0.96  (gate >= 0.50)
  Robust depth sigma     114.2  (gate >= 7.0)
  Odd/even z-score       0.36  (consistent)
  Secondary eclipse      detected, phase 0.49, 6.5 sigma
```

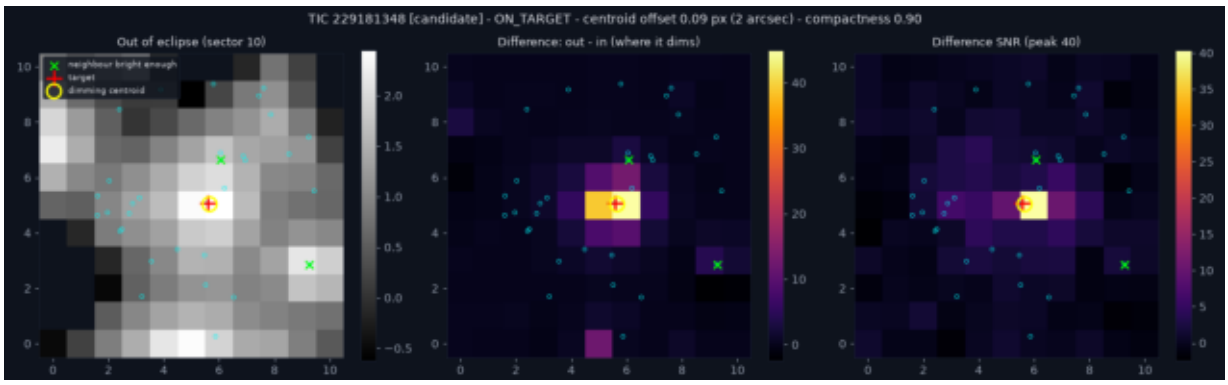
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 10    light-curve dip in this sector: 122.1 sigma    t0 refined by 0.0025 phase
  Difference-image peak SNR   40.39   (gate >= 5.0)      Compactness   0.90   (gate >= 0.50)
  Dimming centroid offset     0.09 px (2 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 229181349
    1.65 px away, dTmag +2.16 - farther from the dimming than the target
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No caveats - clean detection on every check.